

LQR with Quaternions and Quadrotor Control

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- Optimization with Quaternions
- Quaternion differentiation
- Tangent space representation

Naive Linearization Problem

State:

$$x = \begin{bmatrix} p \\ q \\ v \\ \omega \end{bmatrix}$$

If we linearize directly:

$$x_{k+1} = f(x_k, u_k)$$

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Result:

Linearized system becomes **uncontrollable**

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Instead of 4-parameter quaternion, use 3-parameter local representation.
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$$\delta q \in \mathbb{R}^3$$

represent small rotation in tangent space.

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Reduced state:

$$\delta x = \begin{bmatrix} \delta p \\ \delta \theta \\ \delta v \\ \delta \omega \end{bmatrix}$$

Reduced Jacobians

Given nonlinear system:

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Compute full Jacobians:

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Project quaternion part:

$$F_r = EFE^T$$

$$B_r = EB$$

Now system is controllable.

Define rotation from body frame B to reference frame R :

$$q_e = q_R \otimes q_B^{-1}$$

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Small-angle approximation:

$$\delta\theta \approx 2 \operatorname{vec}(q_e)$$

State Definition

State:

$$x = \begin{bmatrix} p \in \mathbb{R}^3 \\ q \in \mathbb{S}^3 \\ v \in \mathbb{R}^3 \\ \omega \in \mathbb{R}^3 \end{bmatrix}$$

Control:

$$u = \begin{bmatrix} T \\ \tau \end{bmatrix}$$

Position:

$$\dot{p} = v$$

Quaternion:

$$\dot{q} = \frac{1}{2}\Omega(\omega)q$$

$$\Omega(\omega) = \begin{bmatrix} 0 & -\omega^T \\ \omega & -[\omega]_{\times} \end{bmatrix}$$

Newton's Law:

$$m\dot{v} = mg + R(q)F_B$$

Thrust in body frame:

$$F_B = \begin{bmatrix} 0 \\ 0 \\ T \end{bmatrix}$$

Translation Dynamics

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Important:

Need rotation:

$R(q)$ = rotation matrix from quaternion

Euler's equation:

$$I\dot{\omega} + \omega \times I\omega = \tau$$

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or

$$\dot{\omega} = I^{-1}(\tau - \omega \times I\omega)$$

Linearized System

After reduction:

$$\delta x_{k+1} = A\delta x_k + B\delta u_k$$

Solve Riccati equation:

$$P = A^T P A - A^T P B (R + B^T P B)^{-1} B^T P A + Q$$

Controller:

$$u = u_{ref} - K\delta x$$

Implementation Notes

- Always renormalize quaternion
- Be consistent with error definition
- Choose small-angle representation carefully
- Avoid sign ambiguity of quaternion

Summary

- Naive quaternion linearization fails
- Use tangent space reduction
- Quadrotor = $SE(3)$ system
- LQR works after proper reduction